

Solution to the Matrix Exponential Limit

Based on the problem provided in dailyintegral-limit-limit-1x1-2026-08-08.jpg, we are tasked with evaluating the following limit:

$$\det \left(\lim_{n \rightarrow \infty} \left(I + \frac{A}{n} \right)^n \right)$$

where $A = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$.

To find this value, we can use the definition of the matrix exponential and its determinant properties.

Rigorous Derivation

1. Identify the Matrix Exponential

The expression inside the determinant is the standard limit definition of the matrix exponential. For any square matrix A :

$$\lim_{n \rightarrow \infty} \left(I + \frac{A}{n} \right)^n = e^A$$

Therefore, the expression simplifies to finding $\det(e^A)$.

2. Apply Jacobi's Formula

A fundamental property of the matrix exponential (often derived via Jacobi's formula) states that the determinant of a matrix exponential is equal to the exponential of the trace of that matrix:

$$\det(e^A) = e^{\text{Tr}(A)}$$

where $\text{Tr}(A)$ is the trace of matrix A , defined as the sum of its main diagonal elements.

3. Calculate the Trace of A

Given our matrix:

$$A = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$$

The main diagonal entries are both 0. Thus, the trace is:

$$\text{Tr}(A) = 0 + 0 = 0$$

4. Evaluate the Determinant

Substituting the trace back into the identity from Step 2:

$$\det(e^A) = e^0 = 1$$

Alternative Approach: Explicit Calculation

Alternatively, we can compute e^A explicitly. Notice that A is the generator for a rotation matrix. Let us evaluate A^2 :

$$A^2 = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} = \begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix} = -I$$

Using the Taylor series expansion for e^A and separating even and odd powers, we obtain Euler's formula for matrices:

$$e^A = I \cos(1) + A \sin(1) = \begin{pmatrix} \cos(1) & \sin(1) \\ -\sin(1) & \cos(1) \end{pmatrix}$$

Now, calculating the determinant of this resulting matrix:

$$\begin{aligned} \det(e^A) &= (\cos(1))(\cos(1)) - (\sin(1))(-\sin(1)) \\ &= \cos^2(1) + \sin^2(1) \\ &= 1 \end{aligned}$$

Both methods confirm that the limit evaluates perfectly to 1.